```
#Importing pandas library
import pandas as pd
```

```
#Loading the dataframe from a csv file using pandas
df = pd.read_csv("Churn.csv")
df.head()
```

Show hidden output

Exploratory Data Analysis

```
#Exploring the dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3333 entries, 0 to 3332
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Account_Length  3333 non-null   int64
 1   Vmail_Message   3333 non-null   int64
 2   Day_Mins        3333 non-null   float64
 3   Eve_Mins        3333 non-null   float64
 4   Night_Mins      3333 non-null   float64
 5   Intl_Mins       3333 non-null   float64
 6   CustServ_Calls  3333 non-null   int64
 7   Churn           3333 non-null   object
 8   Intl_Plan       3333 non-null   object
 9   Vmail_Plan      3333 non-null   object
 10  Day_Calls       3333 non-null   int64
 11  Day_Charge      3333 non-null   float64
 12  Eve_Calls       3333 non-null   int64
 13  Eve_Charge      3333 non-null   float64
 14  Night_Calls     3333 non-null   int64
 15  Night_Charge    3333 non-null   float64
 16  Intl_Calls      3333 non-null   int64
 17  Intl_Charge     3333 non-null   float64
 18  State           3333 non-null   object
 19  Area_Code       3333 non-null   int64
 20  Phone           3333 non-null   object
dtypes: float64(8), int64(8), object(5)
memory usage: 546.9+ KB
```

```
#Exploring the dataset
df.describe()
```

	Account_Length	Vmail_Message	Day_Mins	Eve_Mins	Night_Mi
count	3333.000000	3333.000000	3333.000000	3333.000000	3333.0000
mean	101.064806	8.099010	179.775098	200.980348	200.8720
std	39.822106	13.688365	54.467389	50.713844	50.5738
min	1.000000	0.000000	0.000000	0.000000	23.2000
25%	74.000000	0.000000	143.700000	166.600000	167.0000
50%	101.000000	0.000000	179.400000	201.400000	201.2000
75%	127.000000	20.000000	216.400000	235.300000	235.3000
max	243.000000	51.000000	350.800000	363.700000	395.0000

```
#Exploring Customer Churn
df["Churn"].value_counts()
```

Churn	count
no	2850
yes	483

dtype: int64

Observation: *Imbalanced classes with non-churners being thrice of the churners*

Summary Statistics for both classes

```
#Subsetting the data to investigate specific features
df_custserv_vmsg = df[['Churn','CustServ_Calls','Vmail_Message']]
df_custserv_vmsg.head()
```

	Churn	CustServ_Calls	Vmail_Message
0	no	1	25
1	no	1	26
2	no	0	0
3	no	2	0
4	no	3	0

```
#Mean for Churners vs Non-Churners
df_custserv_vmsg.groupby(["Churn"]).mean()
```

Churn	CustServ_Calls	Vmail_Message
no	1.449825	8.604561
yes	2.229814	5.115942

```
#Standard deviation for Churners vs Non-Churners
df_custserv_vmsg.groupby(["Churn"]).std()
```

Churn	CustServ_Calls	Vmail_Message
no	1.163883	13.913125
yes	1.853275	11.860138

```
#Churn vs Churners by State
df.groupby("State")["Churn"].value_counts()
```

State	Churn	count
AK	no	49
	yes	3
AL	no	72
	yes	8
AR	no	44
...	...	...
WI	yes	7
WV	no	96
	yes	10
WY	no	68
	yes	9

102 rows × 1 columns

dtype: int64

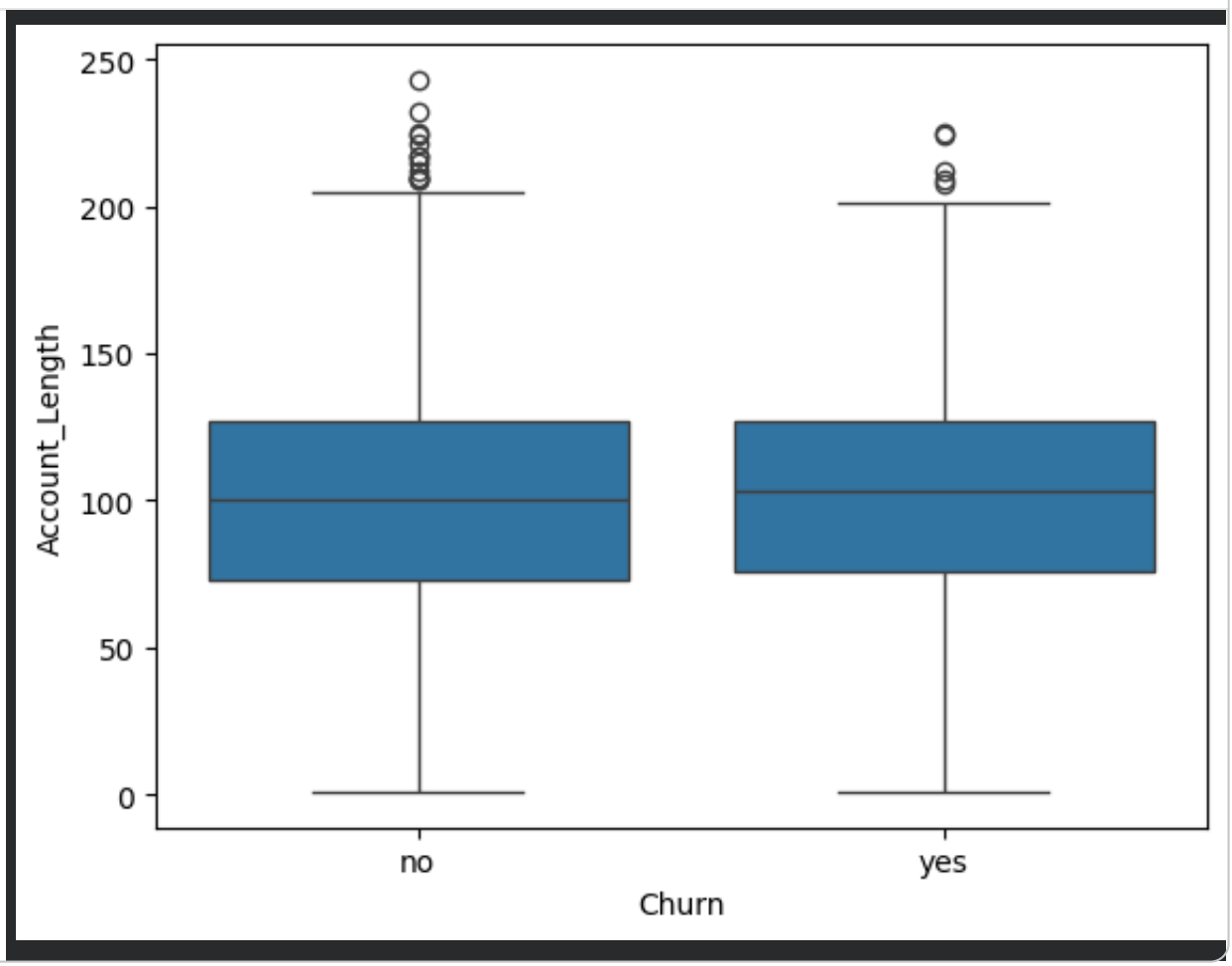
```
#Statewise churn counts comparison
df.groupby("State")["Churn"].value_counts()[["CA","CO","AL"]]
```

State	Churn	count
CA	no	25
	yes	9
CO	no	57
	yes	9
AL	no	72
	yes	8

dtype: int64

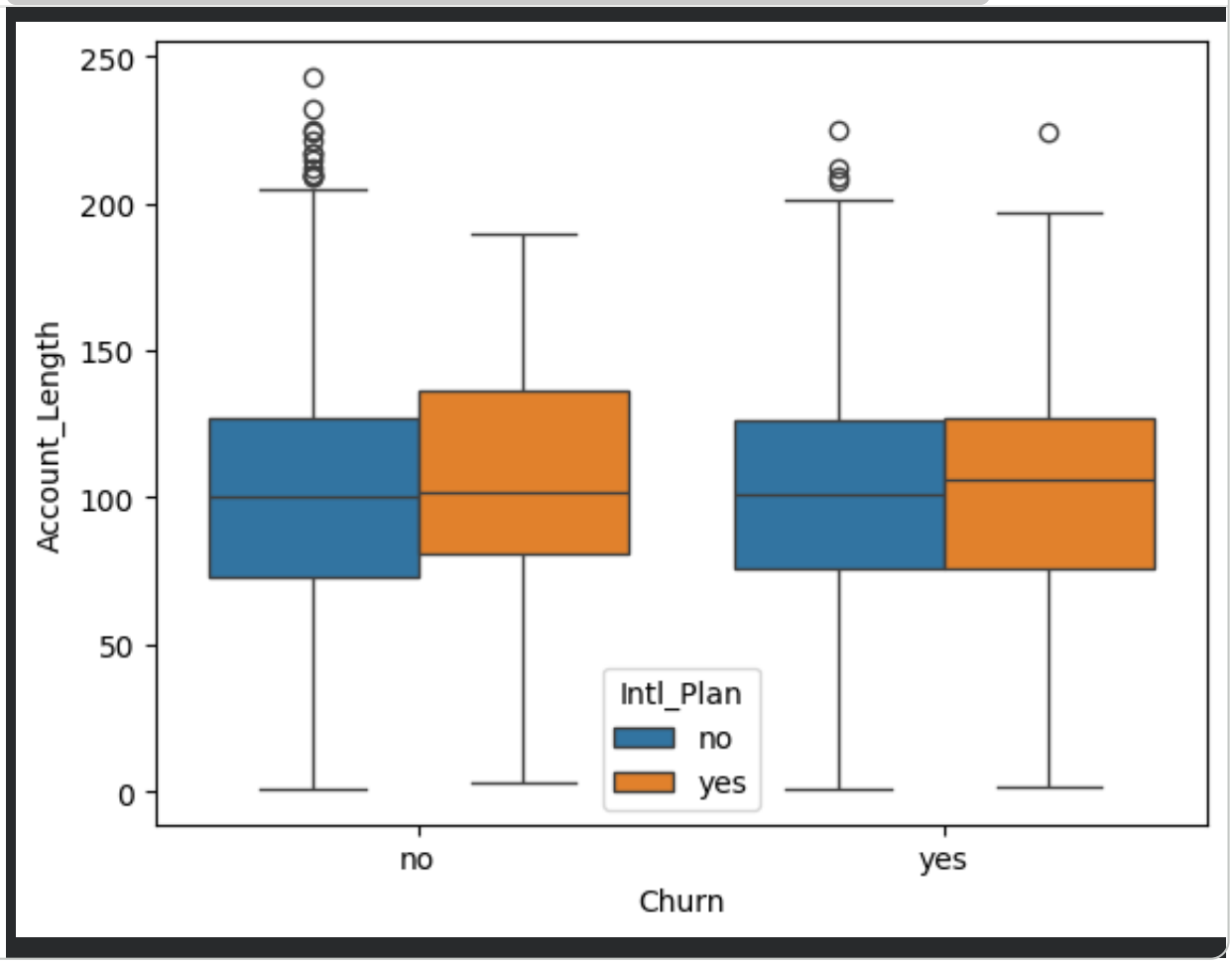
VISUALISING THE DATA

```
#Importing data visualisation libraries - Matplotlib & Seaborn
import matplotlib.pyplot as plt
import seaborn as sns
sns.boxplot(x= 'Churn',
            y= 'Account_Length',
             data = df)
plt.show()
```



Observation:*No difference in account length for Churners vs Non-Churners.*

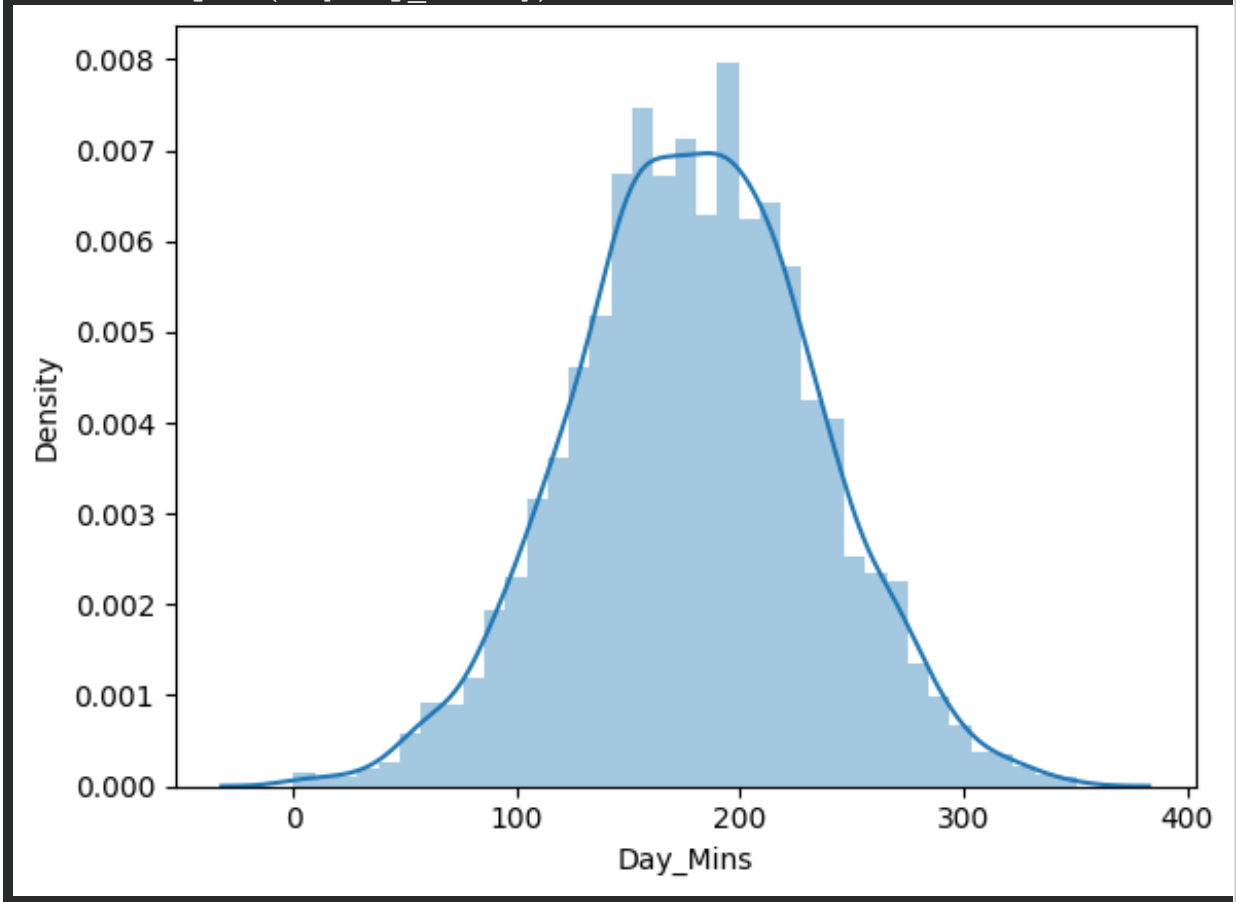
```
# Adding a third variable to understand it's impact on the Churn an
import matplotlib.pyplot as plt
import seaborn as sns
sns.boxplot(x= 'Churn',
            y= 'Account_Length',
            data = df,
            hue='Intl_Plan'
            )
plt.show()
```



Observation:*No impact of International Plan*

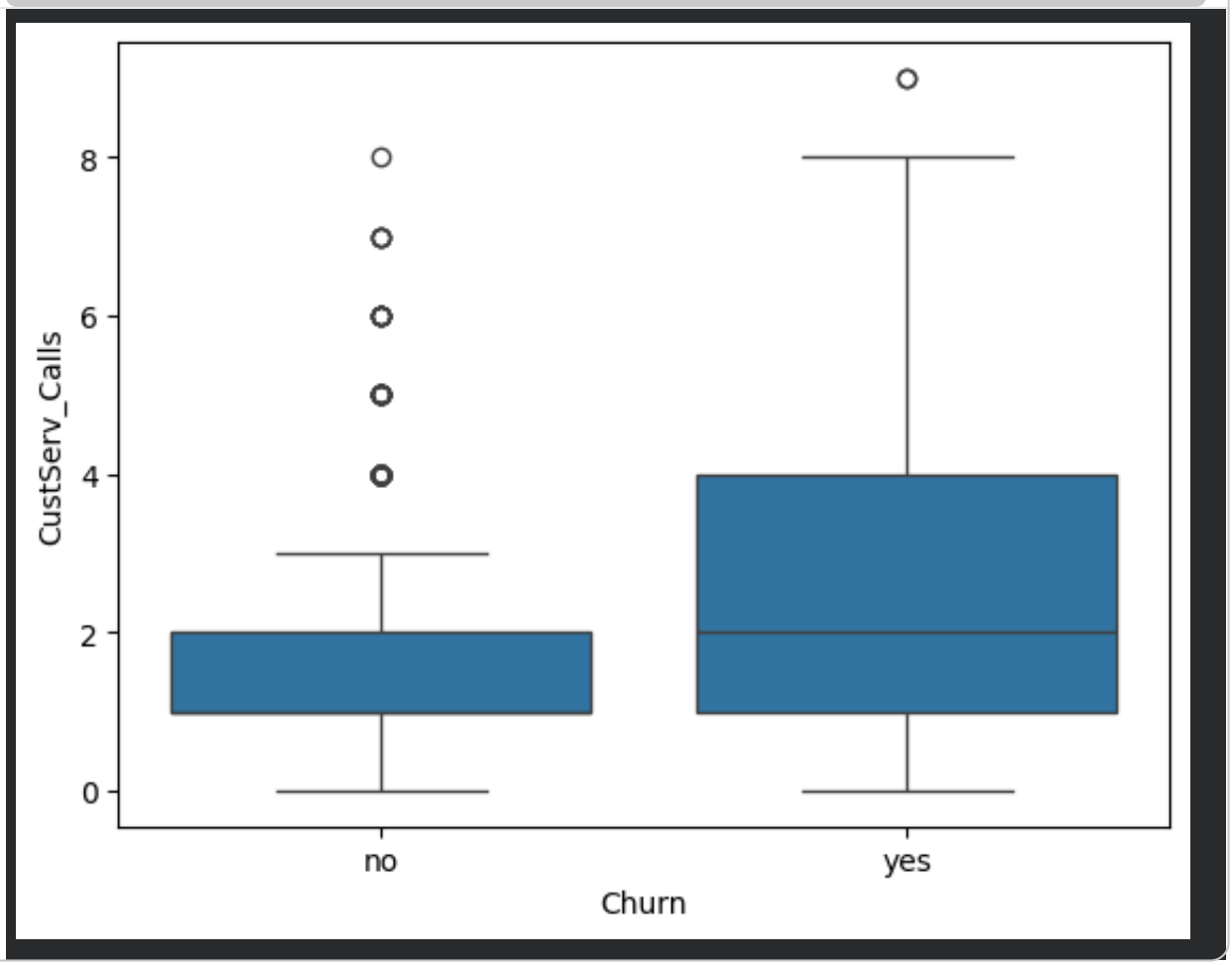
```
#Distribution plot the features(Day_Mins,Eve_Mins, Night_Mins, Intl
sns.distplot(df['Day_Mins'])
plt.show()
```

```
/tmp/ipykernel_6217/3591231309.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v

Please adapt your code to use either `displot` (a figure-level funct
similar flexibility) or `histplot` (an axes-level function for histo

For a guide to updating your code to use the new functions, please s
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['Day_Mins'])
```



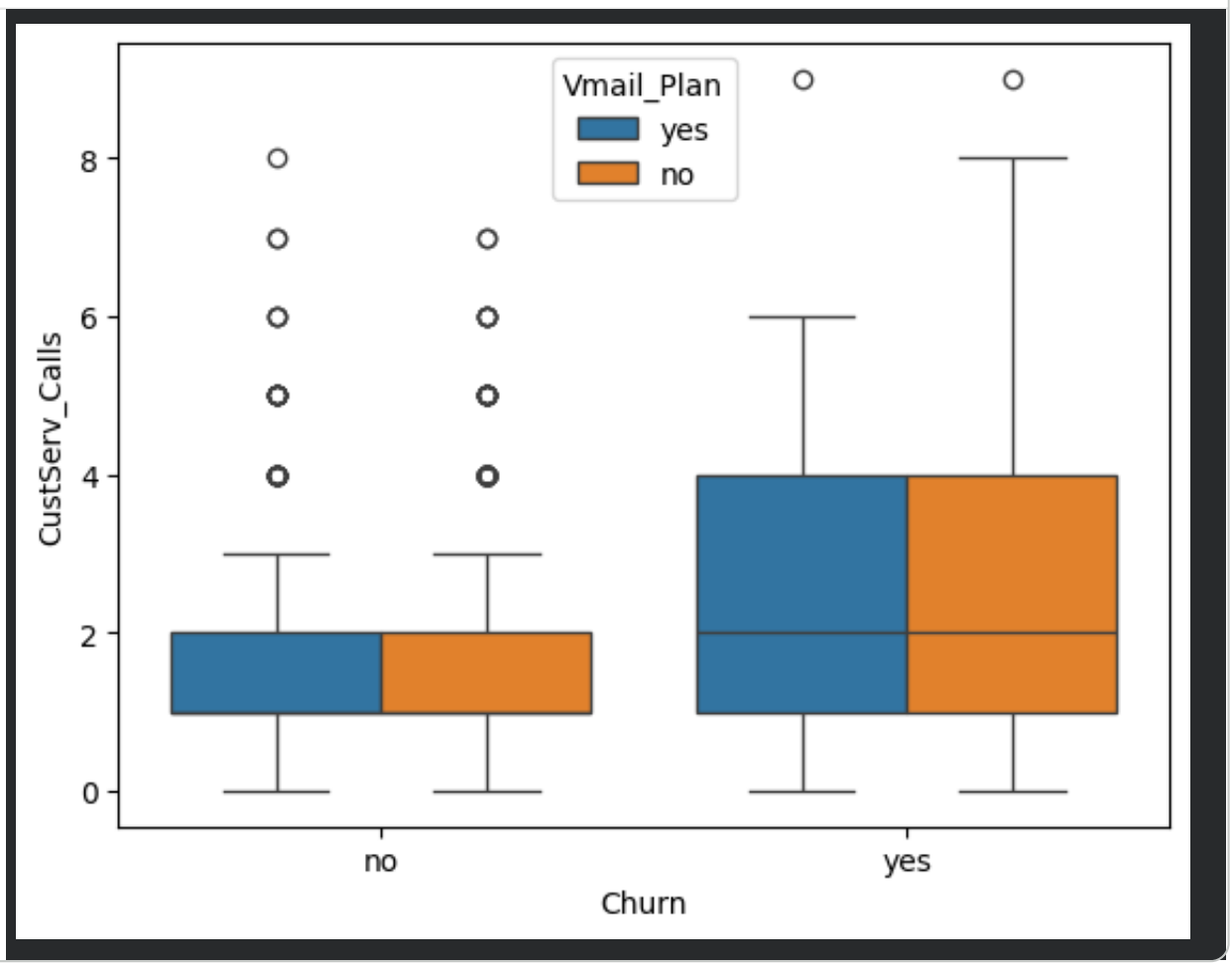
Observation: *All(Day_Mins,Eve_Mins, Night_Mins, Intl_mins) of these features appear to be well approximated by the normal distribution. If this were not the case,we would have to consider applying a feature transformation of some kind.*

```
#Lets investigate the impact of Customer Service Calls on the Churn
sns.boxplot(x='Churn',
            y='CustServ_Calls',
            data=df)
plt.show()
```



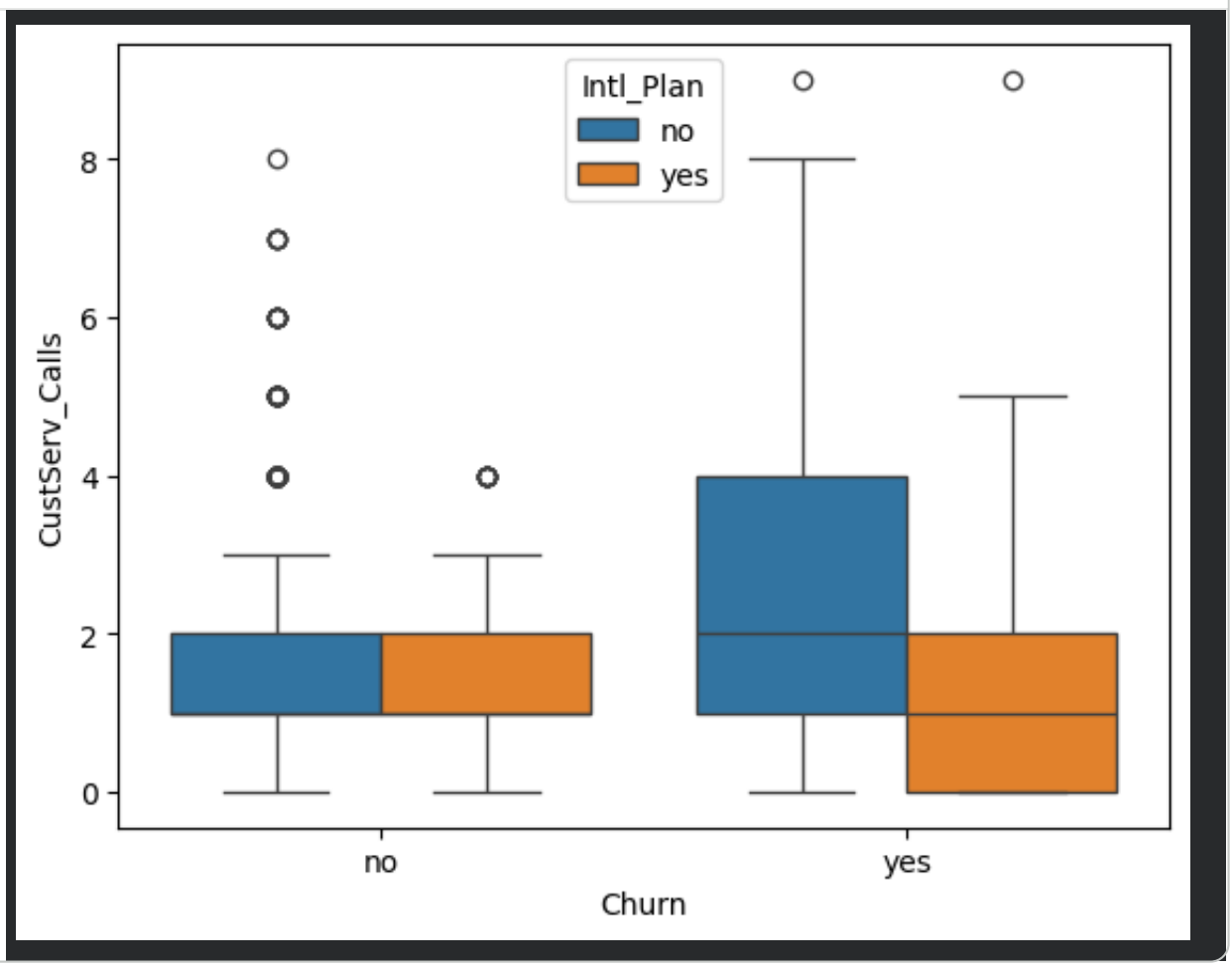
Observation:*# The box plot reveals that the Customer Service Calls feature has a noticeable difference(much higher) for churners compared to non-churners*

```
#Impact of Customer Vmail_Plan on Churn
import matplotlib.pyplot as plt
import seaborn as sns
sns.boxplot(x='Churn',
            y='CustServ_Calls',
            data=df,
            hue='Vmail_Plan')
plt.show()
```



Observation:** *On plotting the box plot, we see that there is not much impact of Vmail_Plan on Churners/Non-churners*

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.boxplot(x='Churn',
            y='CustServ_Calls',
            data=df,
            hue='Intl_Plan')
plt.show()
```



Observations: *Interestingly on plotting the box plot, we see that customers who do churn end up leaving due to more service calls unless they are on a an International plan in which case they make less customer service calls.

Since both Blue and Orange boxes on the right are bigger than on left, it is clear that customers who make service calls churn more but on adding the International plan feature, we see that the box on the extreme right is shifted to a lower value reflecting that customers with Intl plan are making fewer Customer Service calls.

DATA PREPROCESSING

```
#Encoding binary features
df['Churn'] = df['Churn'].replace({'yes':1,'no':0})
df['Vmail_Plan'] = df['Vmail_Plan'].replace({'yes':1,'no':0})
df['Intl_Plan'] = df['Intl_Plan'].replace({'yes':1,'no':0})
df.head()
```

```
/tmp/ipykernel_6217/2454277719.py:2: FutureWarning: Downcasting beha
  df['Churn'] = df['Churn'].replace({'yes':1,'no':0})
/tmp/ipykernel_6217/2454277719.py:3: FutureWarning: Downcasting beha
  df['Vmail_Plan'] = df['Vmail_Plan'].replace({'yes':1,'no':0})
/tmp/ipykernel_6217/2454277719.py:4: FutureWarning: Downcasting beha
  df['Intl_Plan'] = df['Intl_Plan'].replace({'yes':1,'no':0})
```

	Account_Length	Vmail_Message	Day_Mins	Eve_Mins	Night_Mins	Int
0	128	25	265.1	197.4	244.7	
1	107	26	161.6	195.5	254.4	
2	137	0	243.4	121.2	162.6	
3	84	0	299.4	61.9	196.9	
4	75	0	166.7	148.3	186.9	

5 rows × 21 columns

```
#One hot encoding for categorical feature - State
df_state = pd.get_dummies(df['State'])
df_state.head()
```

	AK	AL	AR	AZ	CA	CO	CT	DC	DE	FL	...	S
0	False	False	False	False	False	False	False	False	False	False	...	Fals
1	False	False	False	False	False	False	False	False	False	False	...	Fals
2	False	False	False	False	False	False	False	False	False	False	...	Fals
3	False	False	False	False	False	False	False	False	False	False	...	Fals
4	False	False	False	False	False	False	False	False	False	False	...	Fals

5 rows × 51 columns

```
#Exploring the scale of features -  Intl_calls (Feature Scaling)
df['Intl_Calls'].describe()
```

	Intl_Calls
count	3333.000000
mean	4.479448
std	2.461214
min	0.000000
25%	3.000000
50%	4.000000
75%	6.000000
max	20.000000

dtype: float64

```
#Exploring the scale of features - Night_Mins (Feature Scaling)
df['Night_Mins'].describe()
```

	Night_Mins
count	3333.000000
mean	200.872037
std	50.573847
min	23.200000
25%	167.000000
50%	201.200000
75%	235.300000
max	395.000000

dtype: float64

```
#Feature Engineering - Average Night Calls = diving Night Minutes b
#Creating the new feature
df['Avg_Night_Calls'] = df['Night_Mins']/df['Night_Calls']
df['Avg_Night_Calls'].head()
```

	Avg_Night_Calls
0	2.689011
1	2.469903
2	1.563462
3	2.212360
4	1.544628

dtype: float64

```
#Feature Engineering - Dropping unnecessary features
cols_to_drop =df[['State', 'Area_Code', 'Phone', 'Churn']]
cols_to_drop.head()
df_features = df.drop(cols_to_drop,axis=1)
df_features.head()
```

	Account_Length	Vmail_Message	Day_Mins	Eve_Mins	Night_Mins	Int
0	128	25	265.1	197.4	244.7	
1	107	26	161.6	195.5	254.4	
2	137	0	243.4	121.2	162.6	
3	84	0	299.4	61.9	196.9	
4	75	0	166.7	148.3	186.9	

```
#Feature Scaling
from sklearn.preprocessing import StandardScaler #Importing Standar
df_scaled = StandardScaler().fit_transform(df_features)#Sacle df us
df_for_md = pd.DataFrame(df_scaled,columns=df_features.columns)#Add
df_for_md.head()
```

	Account_Length	Vmail_Message	Day_Mins	Eve_Mins	Night_Mins	Int
0	0.676489	1.234883	1.566767	-0.070610	0.866743	-0.
1	0.149065	1.307948	-0.333738	-0.108080	1.058571	1.
2	0.902529	-0.591760	1.168304	-1.573383	-0.756869	0.
3	-0.428590	-0.591760	2.196596	-2.742865	-0.078551	-1.
4	-0.654629	-0.591760	-0.240090	-1.038932	-0.276311	-0.

MODEL PREDICTION Goal: to predict whether the customer will Churn or not

Target Variable: 'Churn'

Techniques: Supervised Machine Learning

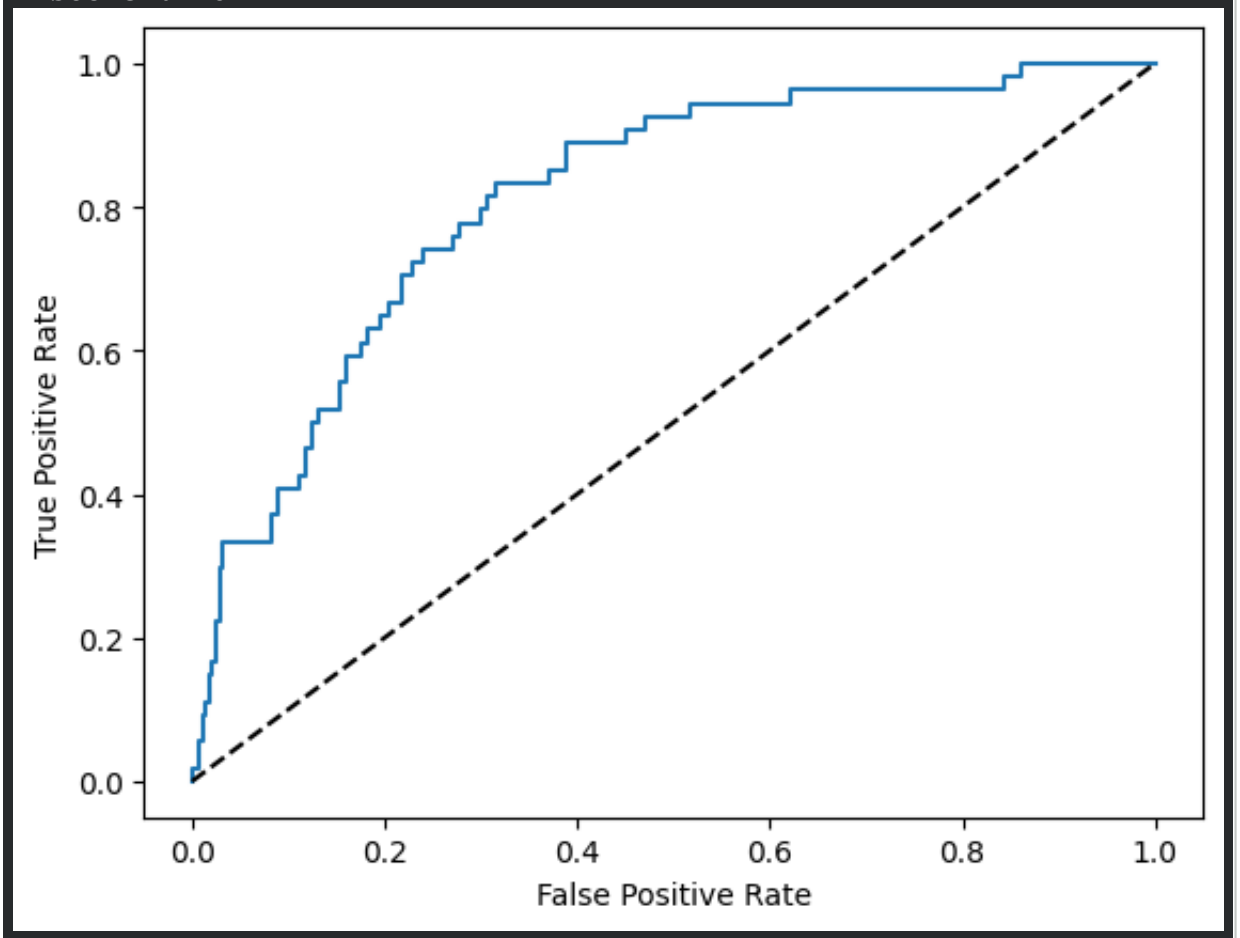
Model Selction: 1.Logistic regression, 2.Decision tree, 3.Support vector machines, 4.Random forests

```
#Importing train test split
from sklearn.model_selection import train_test_split
#Importing LogisticRegression
from sklearn.linear_model import LogisticRegression

#Create training and testing sets
X_train,X_test,y_train,y_test = train_test_split(df_for_md, df['Chu

# Instantiating the classifier
clf=LogisticRegression()

# Fitting into the training data
clf.fit(X_train,y_train)

# Predicting the label for X_test
y_pred = clf.predict(X_test)

#Accuracy Score
```

```
Acc = clf.score(X_test,y_test)
print(f"Accuracy: {Acc:.2f}")

#Precision Score
from sklearn.metrics import precision_score
print(f"Precision: {precision_score(y_test,y_pred):.2f}")

#Recall Score
from sklearn.metrics import recall_score
print(f"Recall: {recall_score(y_test,y_pred):.2f}")

#F1 score
from sklearn.metrics import f1_score
print(f"F1Score:{f1_score(y_test,y_pred):.2f}")

#Other metrics - AOC, ROC

#Generate probabilities
y_pred_prob = clf.predict_proba(X_test)[:,1]

#Importing roc_curve
from sklearn.metrics import roc_curve

#Calculating roc metrics
fpr,tpr,thresholds = roc_curve(y_test,y_pred_prob)

#Plotting the ROC curve
import matplotlib.pyplot as plt
plt.plot(fpr,tpr)

#Add labels and diagonal line(thresholds)
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.plot([0, 1], [0, 1], "k--")
plt.show()

# Importing roc_auc_score
from sklearn.metrics import roc_auc_score

# Printing the AUC(Area Under the Curve)
print(f"AUC :{roc_auc_score(y_test, y_pred_prob):.2f}")
```

```
Accuracy: 0.85
Precision: 0.59
Recall: 0.19
```

```
F1Score:0.28
```




```
AUC :0.81
```

Observations *Class Imbalance and Overfitting beacuse of the high Accuracy and LowF1 score.*

High Accuracy is misleading because Churn being the minority class, the model can predict mostly non - churners and get high accuracy but miss the actual churn cases hence low recall and F1.

```
#Importing Libraries
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

#Creating training and test sets
X_train,X_test,y_train,y_test = train_test_split(df_for_md, df['Chu

#Instantiating the classifier
Clf=DecisionTreeClassifier()

#Fitting into the training data
Clf.fit(X_train,y_train)

#Predicting the label for X_test
y_pred = clf.predict(X_test)

Acc = clf.score(X_test,y_test) #Accuracy Score
print(f"Accuracy: {Acc:.2f}")

from sklearn.metrics import precision_score
print(f"Precision: {precision_score(y_test,y_pred):.2f}") #Precisio

from sklearn.metrics import recall_score
print(f"Recall: {recall_score(y_test,y_pred):.2f}") #Recall

from sklearn.metrics import f1_score
print(f"F1Score:{f1_score(y_test,y_pred):.2f}")#F1 score
```

```
Accuracy: 0.86
Precision: 0.61
Recall: 0.19
F1Score:0.29
```

```
#Importing Libraries
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

#Creating training and test sets
X_train,X_test,y_train,y_test = train_test_split(df_for_md, df['Chu

#Instantiating the classifier
Clf=SVC()

#Fitting into the training data
Clf.fit(X_train,y_train)

#Predicting the label for X_test
y_pred = clf.predict(X_test)

Acc = clf.score(X_test,y_test) #Accuracy Score
print(f"Accuracy: {Acc:.2f}")

from sklearn.metrics import precision_score
print(f"Precision: {precision_score(y_test,y_pred):.2f}") #Precisio

from sklearn.metrics import recall_score
print(f"Recall: {recall_score(y_test,y_pred):.2f}") #Recall

from sklearn.metrics import f1_score
print(f"F1Score:{f1_score(y_test,y_pred):.2f}")#F1 score
```

```
Accuracy: 0.86
Precision: 0.61
Recall: 0.19
F1Score:0.29
```

```
#Importing Libraries
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

#Creating training and test sets
X_train,X_test,y_train,y_test = train_test_split(df_for_md, df['Chu

#Instantiating the classifier
Clf=RandomForestClassifier()

#Fitting into the training data
Clf.fit(X_train,y_train)

#Predicting the label for X_test
y_pred = clf.predict(X_test)

Acc = clf.score(X_test,y_test) #Accuracy Score
print(f"Accuracy: {Acc:.2f}")

from sklearn.metrics import precision_score
print(f"Precision: {precision_score(y_test,y_pred):.2f}") #Precisio

from sklearn.metrics import recall_score
print(f"Recall: {recall_score(y_test,y_pred):.2f}") #Recall

from sklearn.metrics import f1_score
print(f"F1Score:{f1_score(y_test,y_pred):.2f}")#F1 score
```

```
Accuracy: 0.86
Precision: 0.61
Recall: 0.19
F1Score:0.29
```

MODEL TUNING - RandomizedSearchCV

```
# Importing RandomizedSearchCV
from sklearn.model_selection import RandomizedSearchCV
from sklearn.ensemble import RandomForestClassifier
from scipy.stats import randint

# Creating the hyperparameter grid
param_grid = {"max_depth": [3, None],
              "max_features": randint(1, 11),
              "bootstrap": [True, False],
              "criterion": ["gini", "entropy"]}


#Instintiate the classifier
clf = RandomForestClassifier()

# Calling GridSearchCV
grid_search = RandomizedSearchCV(clf,param_grid,cv=3)

# Fitting the model
grid_search.fit(df_for_md, df['Churn'])

# Printing the best hyperparameters
print(grid_search.best_params_)
```

```
{'bootstrap': True, 'criterion': 'entropy', 'max_depth': None, 'max_
```

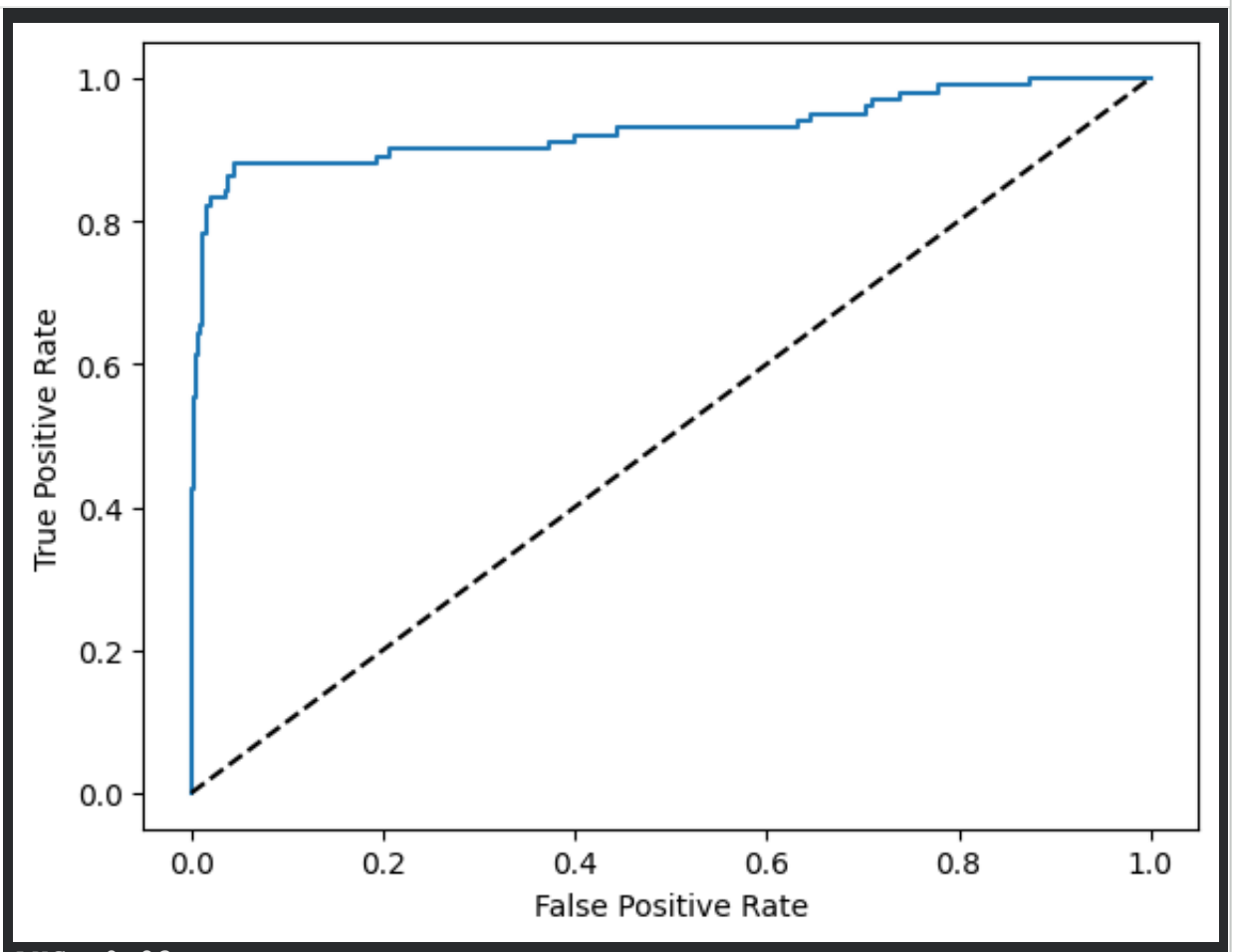
NEW IMPROVED MODEL

```
from sklearn.ensemble import RandomForestClassifier

#Instantiating the classifier with the best hyperparameters
clf = RandomForestClassifier(
    n_estimators=200,
    criterion='entropy',
    max_depth=10,   #To control overfitting
    max_features=9,
    min_samples_split=10,
    min_samples_leaf=5,
    bootstrap=True,
    class_weight='balanced',
    random_state=42
)

#Fitting the training set
clf.fit(X_train, y_train)

#Predicting
y_pred = clf.predict(X_test)


from sklearn.metrics import accuracy_score, precision_score, recall

print(f"Accuracy: {accuracy_score(y_test, y_pred):.2f}")
print(f"Precision: {precision_score(y_test, y_pred):.2f}")
print(f"Recall: {recall_score(y_test, y_pred):.2f}")
print(f"F1 Score: {f1_score(y_test, y_pred):.2f}")
```

```
Accuracy: 0.95
Precision: 0.94
Recall: 0.74
F1 Score: 0.83
```

```
#Other metrics - AOC, ROC

#Generate probabilities
y_pred_prob = clf.predict_proba(X_test)[:,1]

from sklearn.metrics import roc_curve

#Calculate roc metrics
fpr,tpr,thresholds = roc_curve(y_test,y_pred_prob)

#Plot the ROC curve
import matplotlib.pyplot as plt
plt.plot(fpr,tpr)
```

```
#Add labels and diagonal line(thresholds)
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.plot([0, 1], [0, 1], "k--")
plt.show()

# Import roc_auc_score
from sklearn.metrics import roc_auc_score

# Print the AUC(Area Under the Curve)
print(f"AUC :{roc_auc_score(y_test, y_pred_prob):.2f}")
```




```
AUC :0.93
```

Close

```
**OBSERVATION** *The model is drasti
hyperparameters. The original method
None), no class balancing and severa
memorization of training data and po

*To build the better model, we used
balancing and controlled overfitting
min_samples_split=10,min_samples_lea
```

OBSERVATION *The model is drastically improves by tuning the hyperparameters. The original method deafults to Deep tress (max_depth= None), no class balancing and several correlated features which leads to memorization of training data and poor generalisation*

To build the better model, we used tuned parameters, added class balancing and controlled overfitting by (max_depth=10,min_samples_split=10, min_samples_leaf=5).

VISUALISING THE FEATURE IMPORTANCE

```
from sklearn.ensemble import RandomForestClassifier
import numpy as np
import matplotlib.pyplot as plt

#Instintiate the classifier
clf = RandomForestClassifier()

#Fit the model
clf.fit(df_for_md, df['Churn'])

# Calculate feature importances
importances = clf.feature_importances_

# Sort importances
sorted_index = np.argsort(importances)

# Create labels
labels = df_for_md.columns[sorted_index]

# Clear current plot
plt.clf()

# Create plot
```

```
plt.barh(range(df_for_md.shape[1]), importances[sorted_index], tick
plt.show()
```